

AI for optimal control and state preparation in open quantum systems

PhD candidate: Perruzza Maria Elena

PhD supervisor: Massel Francesco

Quantum Systems

EVOLUTION $i\hbar \frac{\partial}{\partial t} |\Psi\rangle = \hat{H} |\Psi\rangle$ Schrödinger equation

Quantum Systems + Environment

EVOLUTION $\frac{d\rho}{dt} = -\frac{i}{\hbar} [H, \rho] + \sum_j \gamma_j \left(L_j \rho L_j^\dagger - \frac{1}{2} \{L_j^\dagger L_j, \rho\} \right)$

Lindblad equation

Accounting for environmental interactions, such as **energy relaxation** and **decoherence**

Agenda



Problems

1. **Non-classical states** preparation
via **AI-based quantum control** and
by accounting for **non-unitary evolution**

2. Mitigating **dissipation** ?

1. **Non-classical states** preparation
via AI-based quantum control and
by accounting for **non-unitary evolution**

Problems

motivation

Gradient Based Models (GBM) problems

unitary bias

primarily optimised for Schrödinger dynamics

vanishing gradients

prone to local minima in high-dimensional systems

lack of robustness

GBM-optimised pulses are highly sensitive to model inaccuracies



1. **Non-classical states** preparation via AI-based quantum control and by accounting for **non-unitary evolution**

Problems

motivation

AI-driven control: overcoming GBM limitations

beyond unitary dynamics

direct integration of the Master Equation

AI can explore the global control landscape, avoiding sub-optimal traps in complex, high-dimensional systems

robustness

AI can learn a control policy that is robust to fluctuation



Example

Q. Rumman Rahman,¹ Igor Kladarić,^{2,3} Max-Emanuel Kern,^{2,3}
Lukáš Lachman,^{4,5} Yiwen Chu,^{2,3} Radim Filip,^{5,*} and Matteo Fadel^{2,3,†}

¹Department of Physics, Yale University, New Haven, 06511, CT, USA

²Department of Physics, ETH Zürich, 8093 Zürich, Switzerland

³Quantum Center, ETH Zürich, 8093 Zürich, Switzerland

⁴Laboratoire Kastler Brossel, Sorbonne Université, CNRS,

ENS-Université PSL, Collège de France, 4 Place Jussieu, 75005 Paris, France

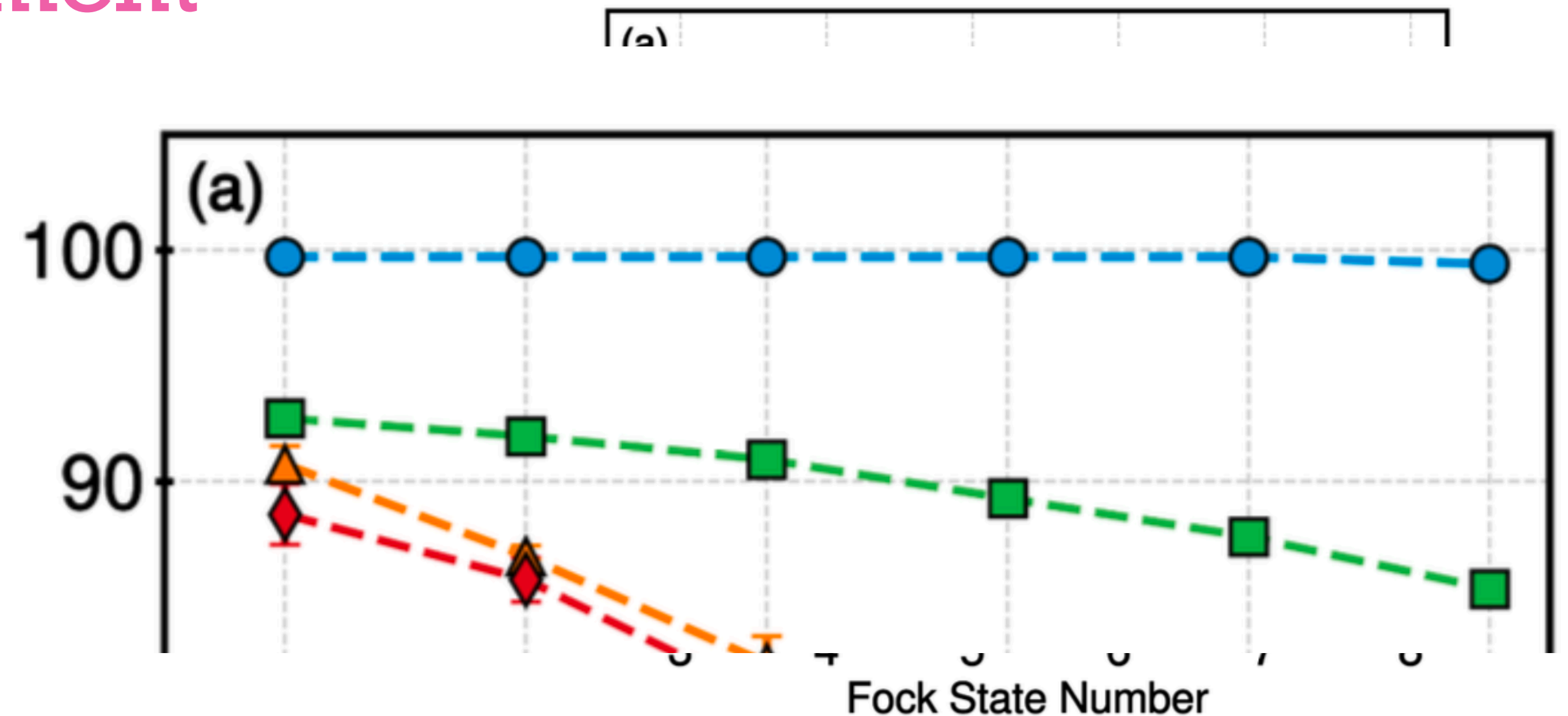
⁵Department of Optics, Palacký University, 17. listopadu 12, 771 46 Olomouc, Czech Republic

They used GRAPE (unitary) for the drive optimisation during the experiment

QUANTUM SYSTEM:
acoustic resonator + QUBIT

**FOCK-STATE PREPARATION
EXPERIMENT**

$$H = -\frac{\sigma_z}{2}(\omega_q - \omega_a) + g(a^\dagger \sigma_- + a \sigma_+) + H_d$$



Fock states are eigen-vectors of the harmonic oscillator Hamiltonian

Phys. Rev. Lett. 134, 180801 (2025)

Methods

AI paradigms for quantum control

Supervised Learning (SL) & Regression
training on large datasets

Deep Reinforcement Learning (DRL)
Agent-Environment interaction loop

Evolutionary & Genetic Algorithms
Genetic Algorithms (GA) or Particle Swarm Optimisation

Physics-Informed Neural Networks (PINNs)
integrating the Schrödinger or Lindblad equations into the Neural Network's loss function

Methods

AI paradigms for quantum control

Supervised Learning (SL) & Regression
training on large datasets

Deep Reinforcement Learning (DRL)
Agent-Environment interaction loop

Evolutionary & Genetic Algorithms
Genetic Algorithms (GA) or Particle Swarm Optimisation

Physics-Informed Neural Networks (PINNs)
integrating the Schrödinger or Lindblad equations into the Neural Network's loss function

Methods

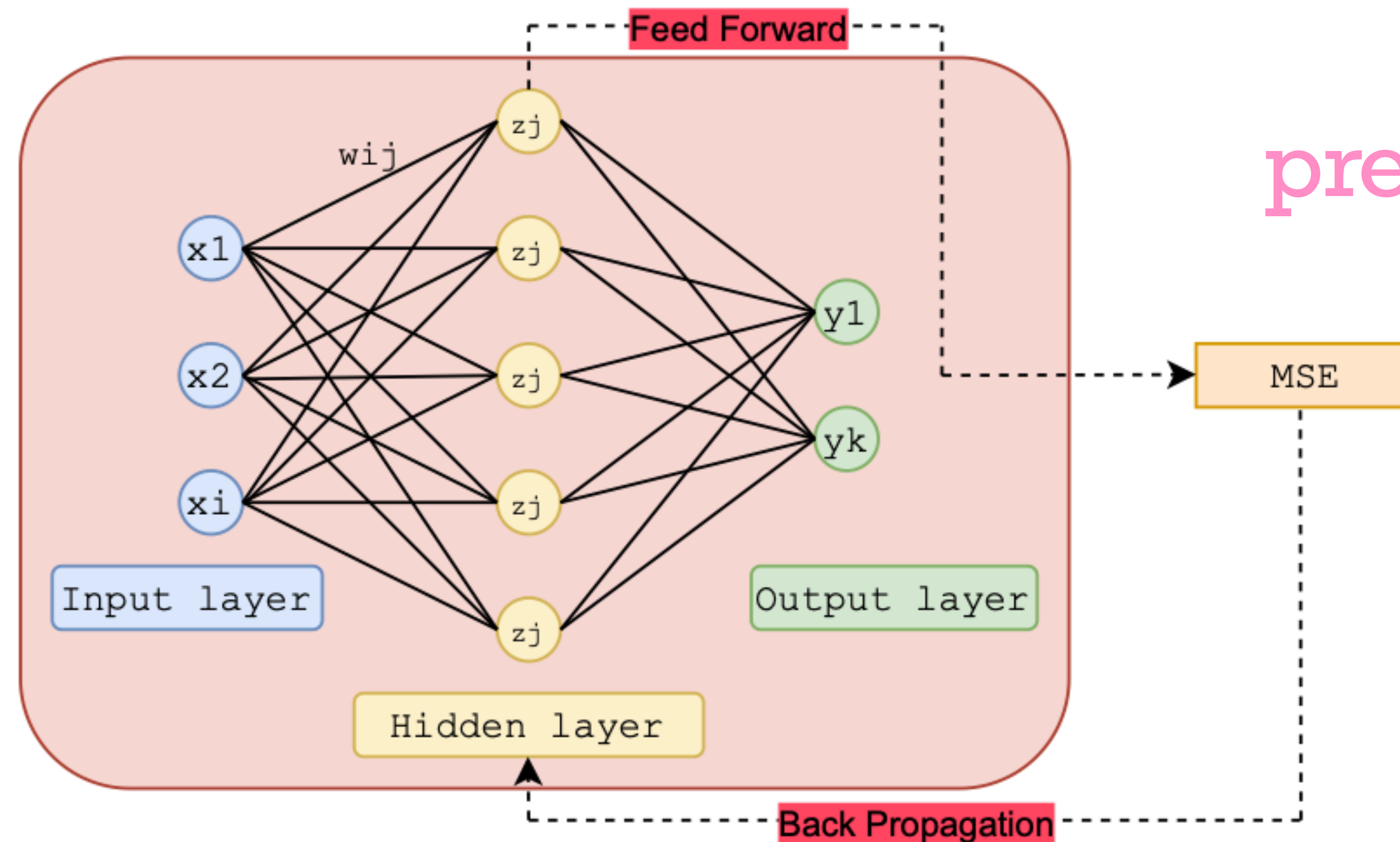
AI paradigms for quantum control

Supervised Learning (SL) & Regression training on large datasets

training phase

the algorithm learns
the mapping
between
Input and Output

$$w_{\text{new}} = w_{\text{old}} - \alpha \underbrace{\frac{dJ}{dw}}_{\text{Gradient}}$$



prediction phase

the algorithm uses
the learned
mapping to
predict the output

$$\hat{y}(t) = f_{\theta}(x(t))$$

AI paradigms for quantum control

Methods

Supervised Learning (SL) & Regression

$$w_{\text{new}} = w_{\text{old}} - \alpha \underbrace{\frac{dJ}{dw}}_{\text{Gradient}}$$

SL is also a **gradient-based** model..
but there is a difference with, for
example, GRAPE

GRAPE

Physical gradient

It is calculated via the **Schrödinger**
or **Lindblad** equations;
It requires solving the system's physics
at every iteration

Supervised Learning

Neural Network weights gradient

It uses **back-propagation** to minimise the error
between predicted output and the real output

AI paradigms for quantum control

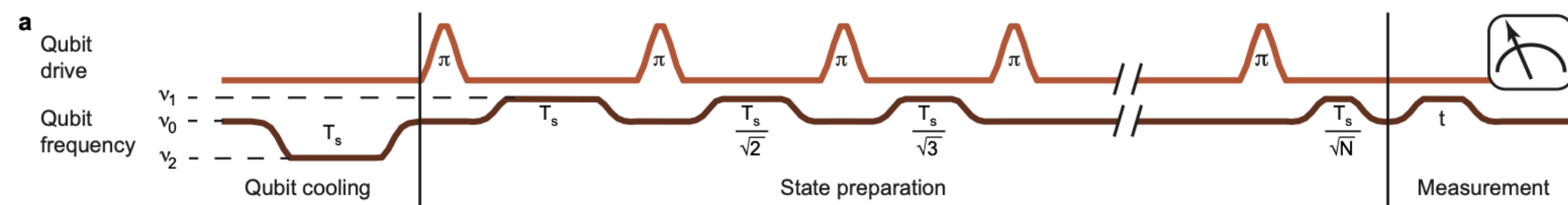
Methods

Supervised Learning (SL) & Regression

Creation and control of multi-phonon Fock states in a bulk acoustic-wave resonator

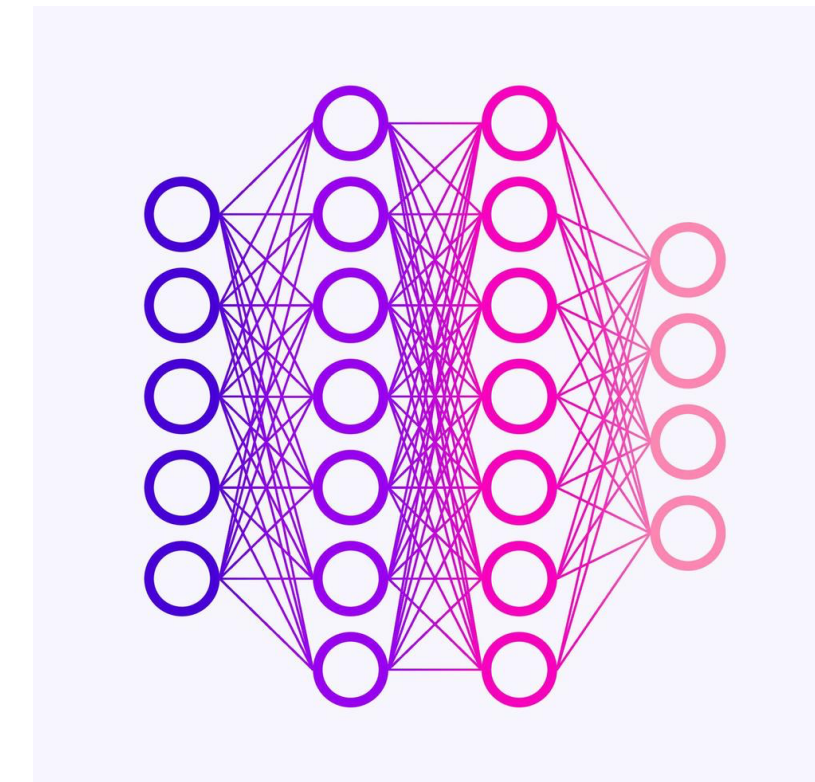
Yiwen Chu^{1,2*}, Prashanta Kharel^{1,2}, Taekwan Yoon^{1,2}, Luigi Frunzio^{1,2}, Peter T. Rakich^{1,2} & Robert J. Schoelkopf^{1,2*}

QUANTUM SYSTEM: acoustic resonator + QUBIT



Nature **563**, 666–670 (2018)

target state
+
fidelity



drive
durations
 ω_d

$$\mathcal{H} = \frac{\Delta_0}{2} \sigma_z + g (\sigma_+ b + \sigma_- b^\dagger) + \Omega_R \cos(\omega_d t) \sigma_x$$

$$F(\rho, \sigma) = \left(\text{tr} \sqrt{\sqrt{\rho} \sigma \sqrt{\rho}} \right)^2$$

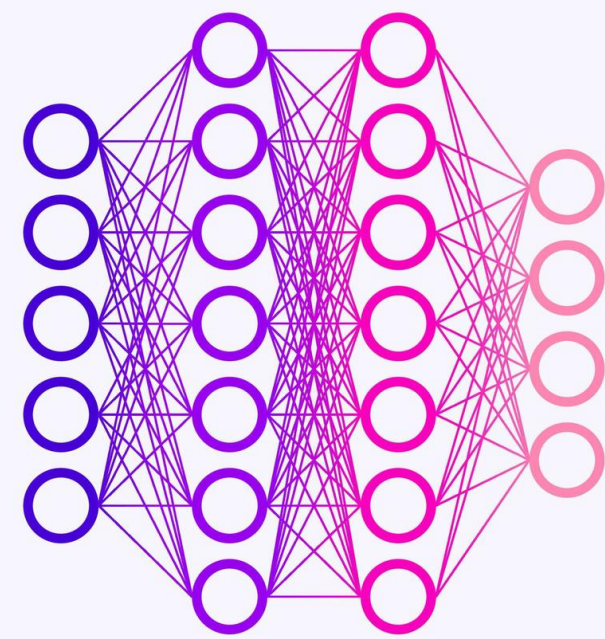
Training phase using a synthetic dataset

AI paradigms for quantum control

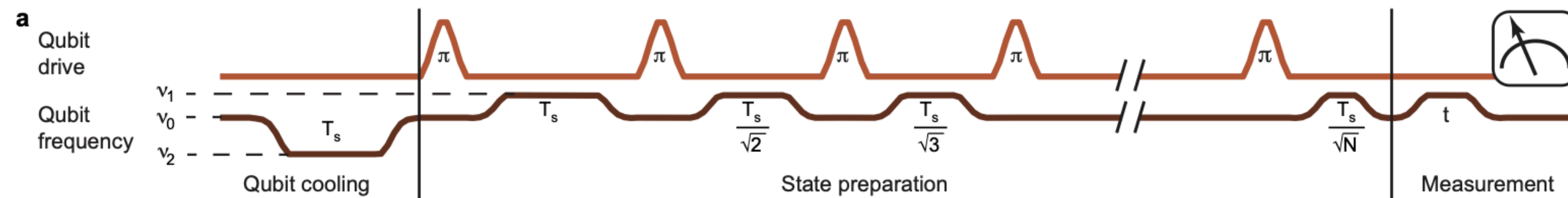
Results

Supervised Learning (SL) & Regression

target state
+
fidelity

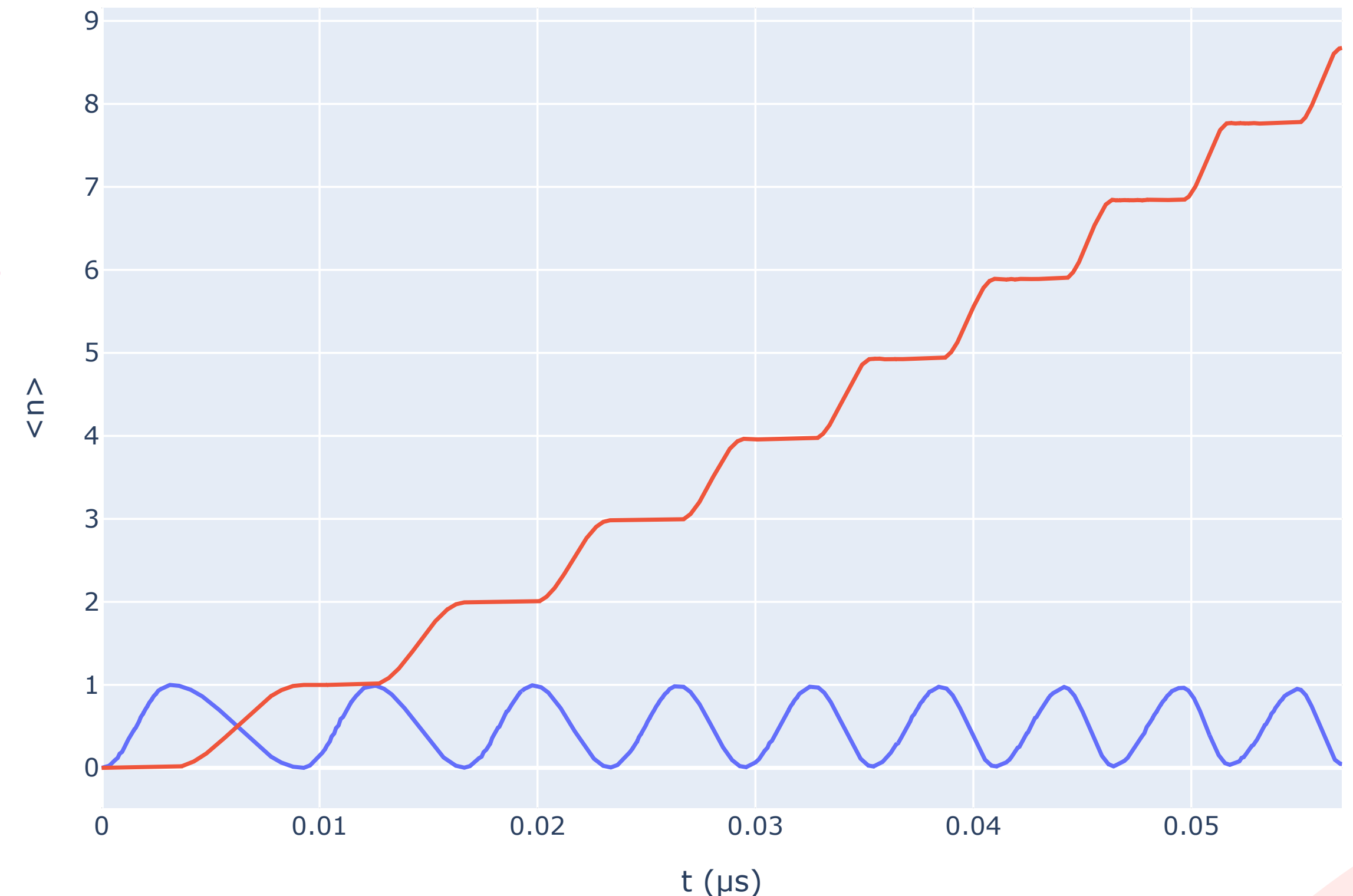


drive
durations
 ω_d



Nature **563**, 666–670 (2018)

$$T_n = T_s \left(1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \dots + \frac{1}{\sqrt{n}} \right)$$



Davide Stirpe, USN

Methods

AI paradigms for quantum control

Supervised Learning (SL) & Regression
training on large datasets

Deep Reinforcement Learning (DRL)
Agent-Environment interaction loop

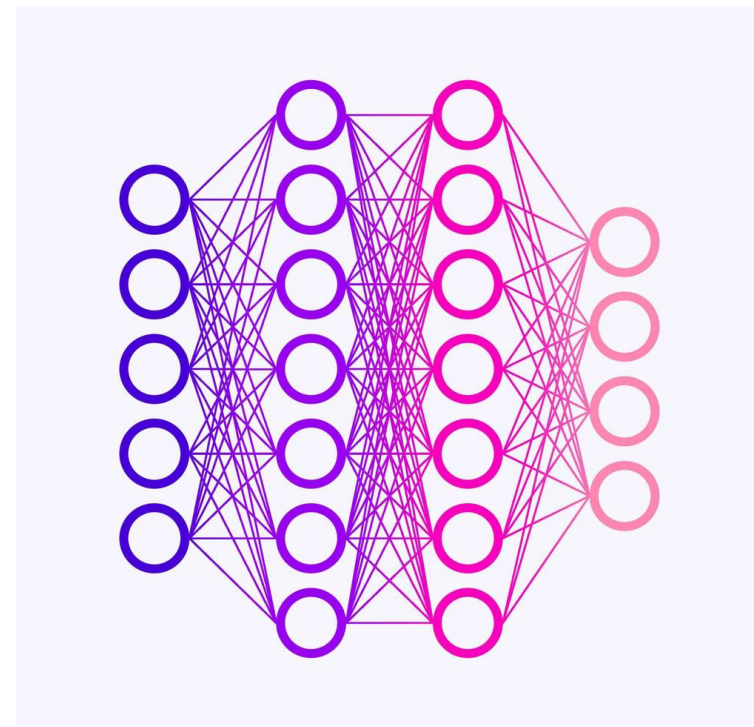
Evolutionary & Genetic Algorithms
Genetic Algorithms (GA) or Particle Swarm Optimisation

Physics-Informed Neural Networks (PINNs)
integrating the Schrödinger or Lindblad equations into the Neural Network's loss function

AI paradigms for quantum control

Methods

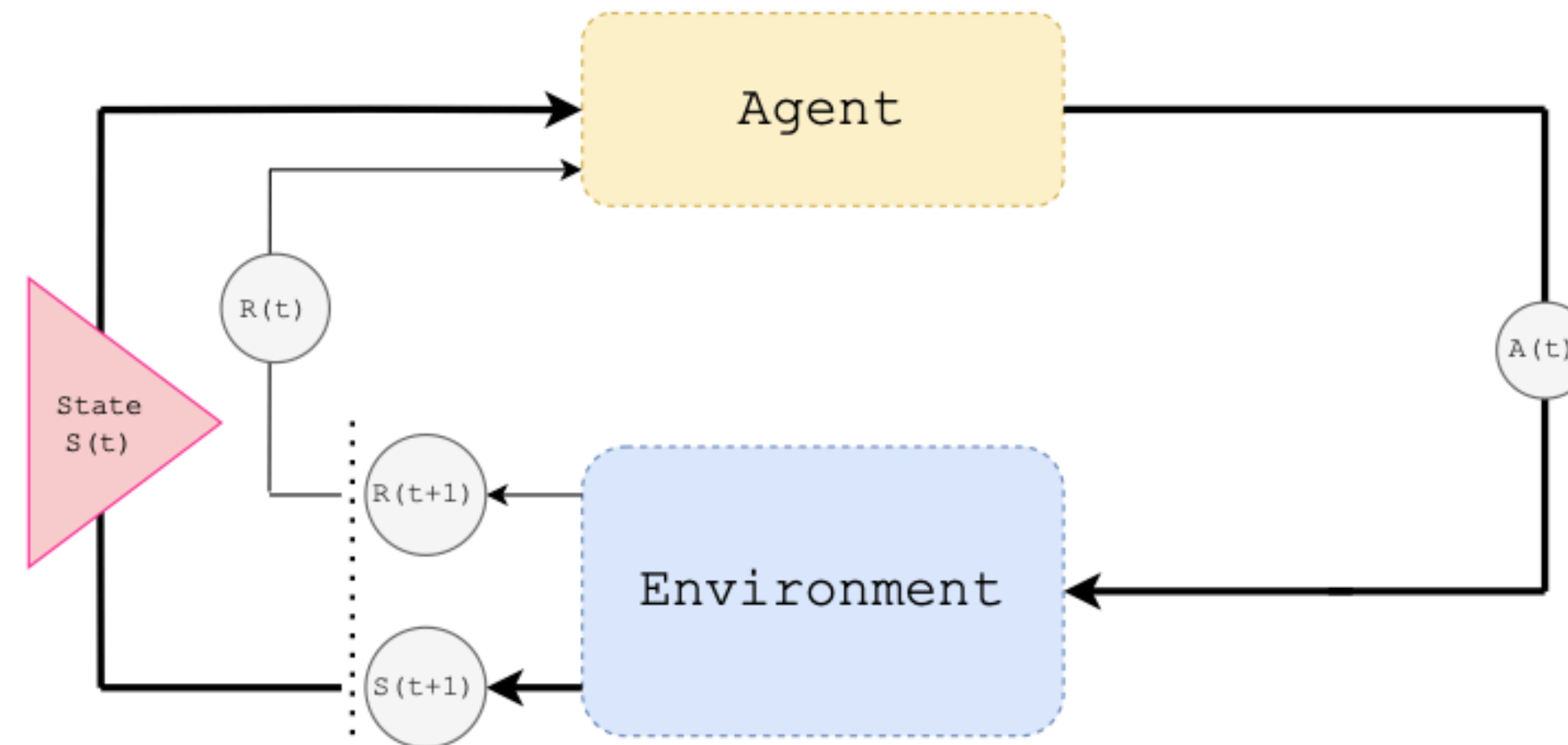
Deep Reinforcement Learning (DRL)
Agent-Environment interaction loop



RL agent

$\pi_{\theta}(a|s)$ (Policy)

Policy has parameters θ



RL model-free

$\tau = \{s_0, a_0, s_1, a_1, \dots, s_t, a_t\}$ (Trajectory)

$R(\tau) = \sum_t \gamma^t r_{s_t}$ (Return)

$s \in \mathcal{S}$ (State) $a \in \mathcal{A}$ (Action) $r_s \in \mathbb{R}$ (Reward)

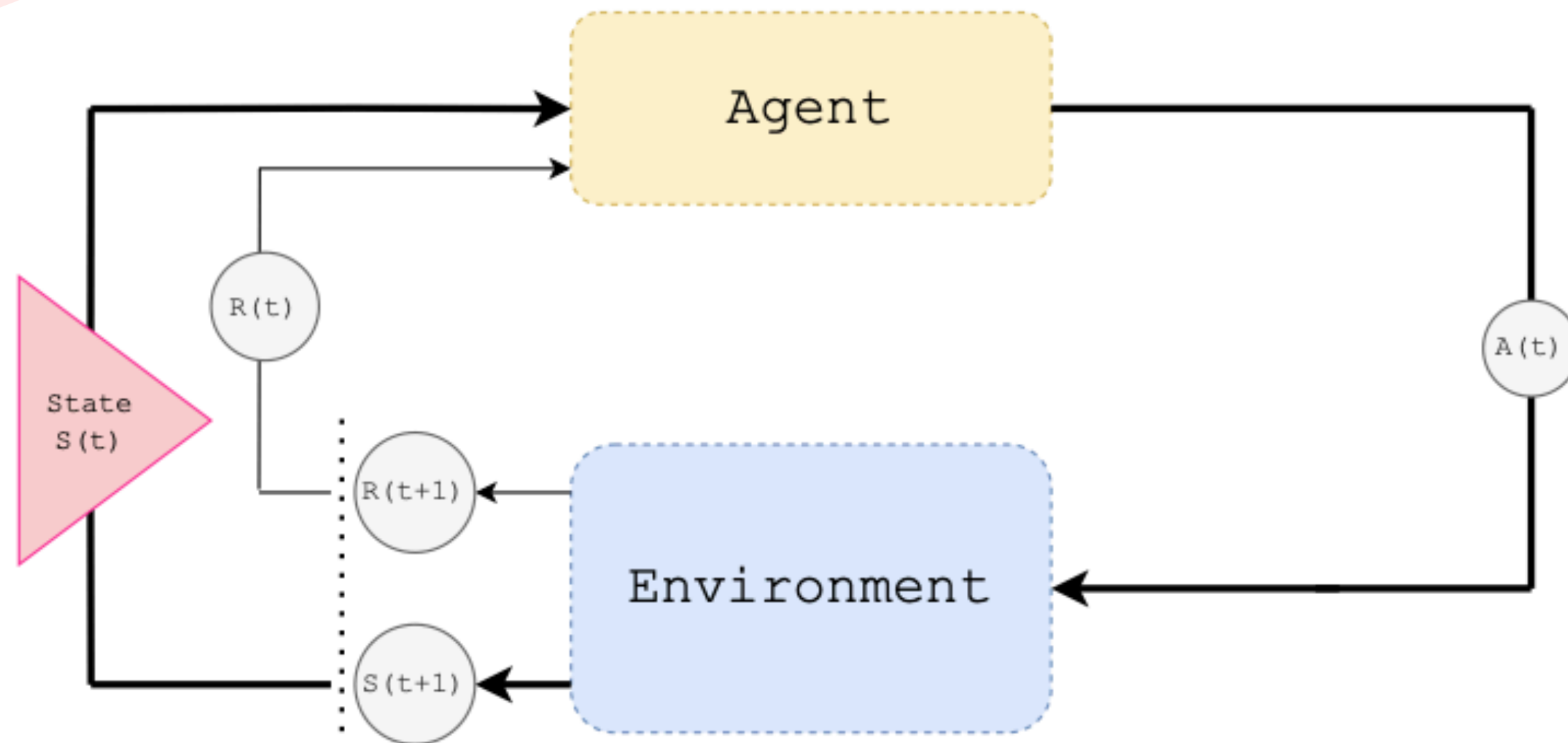
$T(s_{t+1}|s_t, a_t)$ (Transition) $\pi(a|s)$ (Policy)

RL environment

AI paradigms for quantum control

Methods

Deep Reinforcement Learning (DRL) Agent-Environment interaction loop



$\tau = \{s_0, a_0, s_1, a_1, \dots, s_t, a_t\}$ (Trajectory)

$$R(\tau) = \sum_t \gamma^t r_{s_t} \quad (\text{Return})$$

$$J(\pi_\theta) = \mathbb{E}_{\tau \sim (\pi_\theta, T)} [R(\tau)]$$

Diagram showing the components of the equation: Policy (blue box) points to π_θ , Trajectory (red box) points to τ , and Return (green box) points to $R(\tau)$. A Transition Function (orange box) points to the expectation operator $\mathbb{E}$.

Policy Optimization with Gradient Ascent

$$\theta_{t+1} = \theta_t + \alpha \nabla_\theta J(\pi_\theta) |_{\theta_t}$$

Basic Policy Gradient

$$\nabla_\theta J(\pi_\theta) = \mathbb{E}_{\tau \sim (\pi_\theta, T)} \left[\sum_{t=0}^T \nabla_\theta \log \pi_\theta(a_t | s_t) R(\tau) \right]$$

AI paradigms for quantum control

Methods

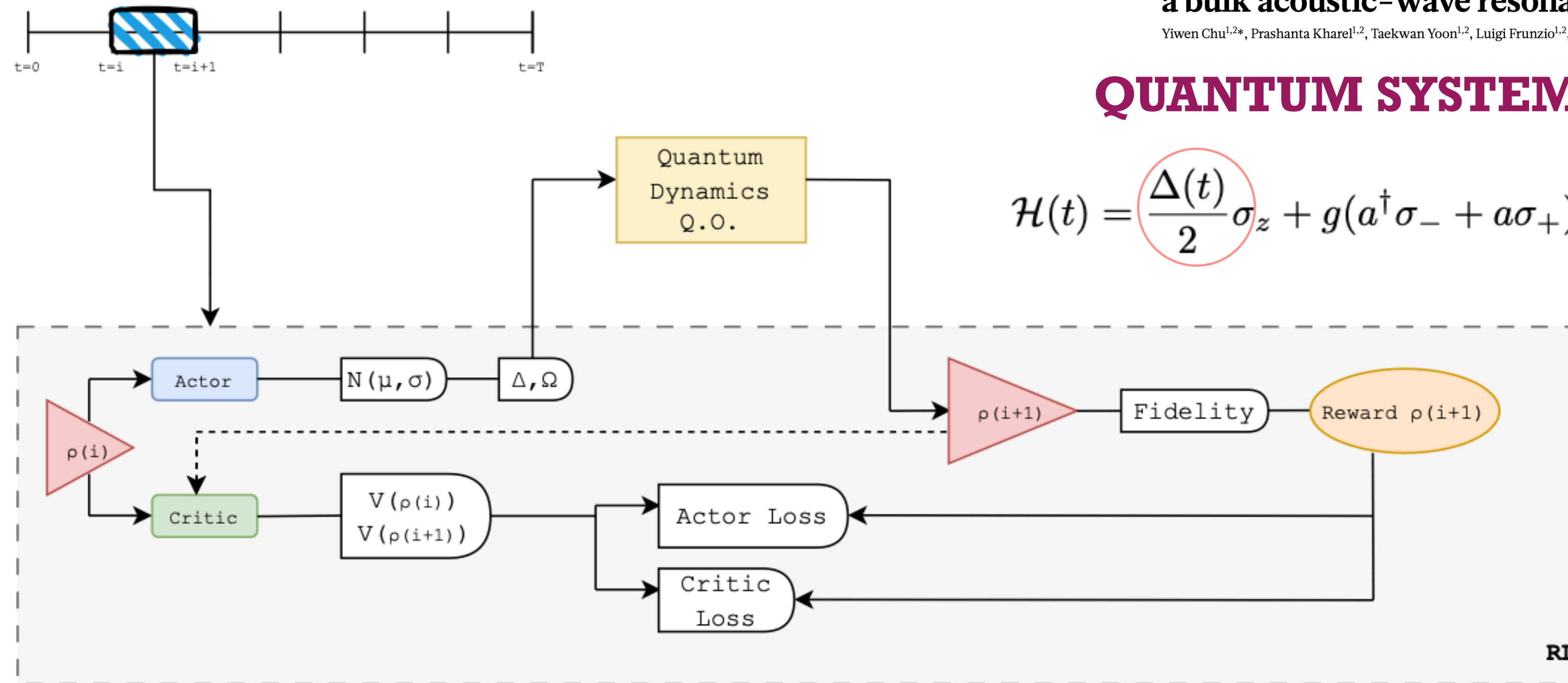
Deep Reinforcement Learning (DRL) Agent-Environment interaction loop

Creation and control of multi-phonon Fock states in a bulk acoustic-wave resonator

Yiwen Chu^{1,2*}, Prashanta Kharel^{1,2}, Taekwan Yoon^{1,2}, Luigi Frunzio^{1,2}, Peter T. Rakich^{1,2} & Robert J. Schoelkopf^{1,2*}

QUANTUM SYSTEM: HBAR + QUBIT

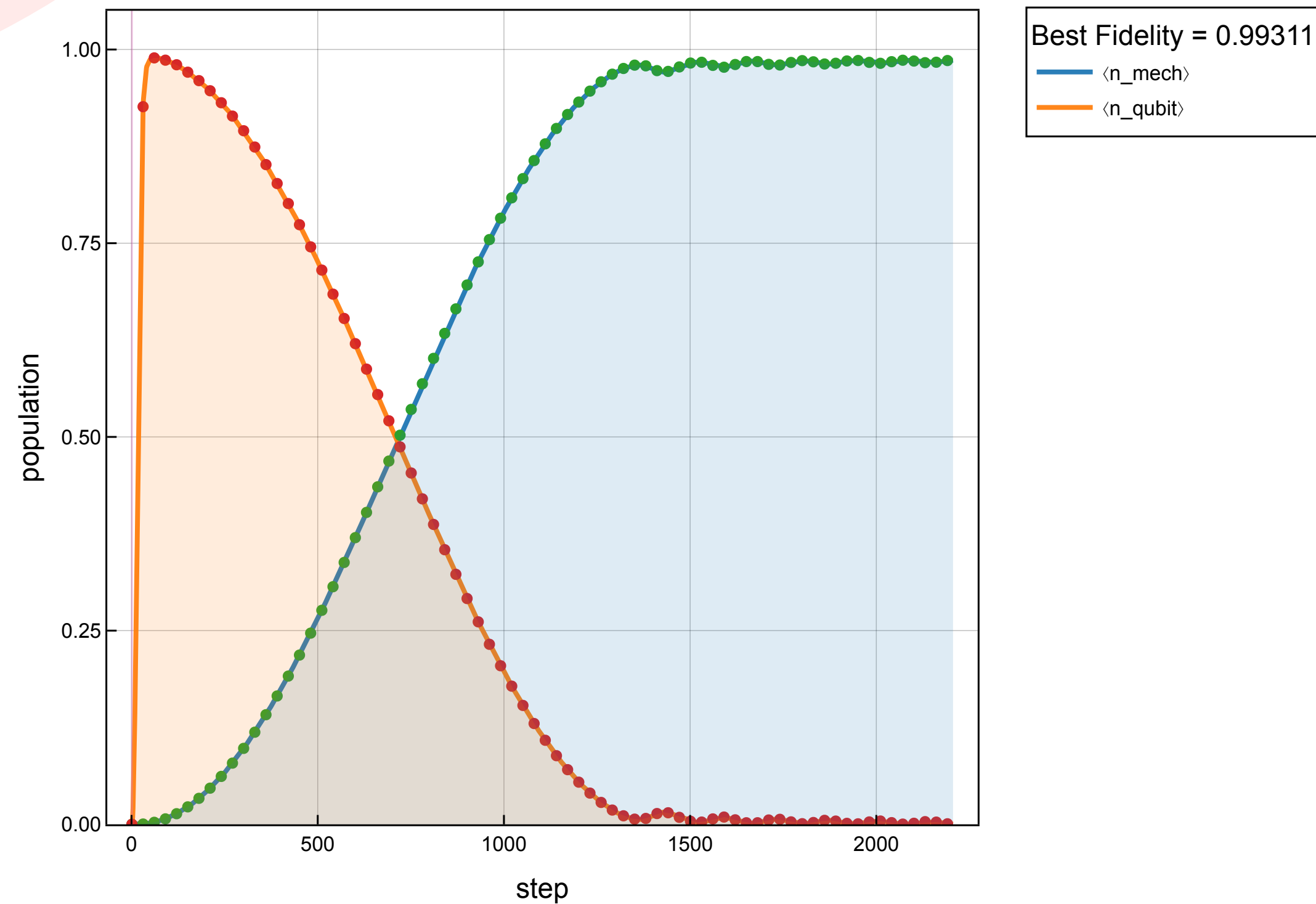
$$\mathcal{H}(t) = \frac{\Delta(t)}{2} \sigma_z + g(a^\dagger \sigma_- + a \sigma_+) + \Omega_R(t) \sigma_x, \quad \Delta = \omega_q - \omega_m$$



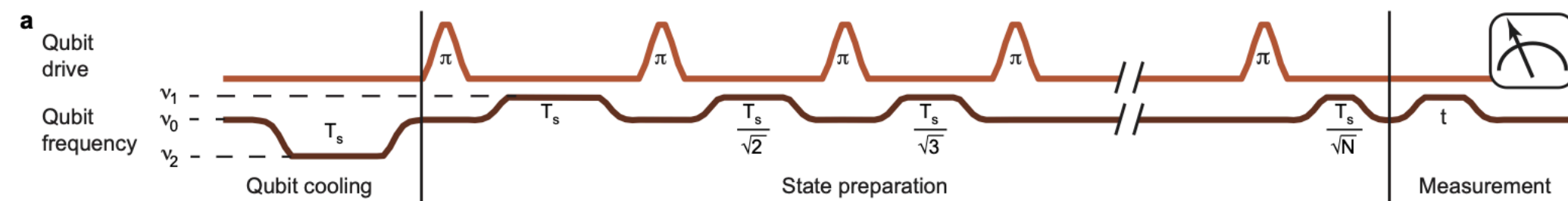
AI paradigms for quantum control

Results

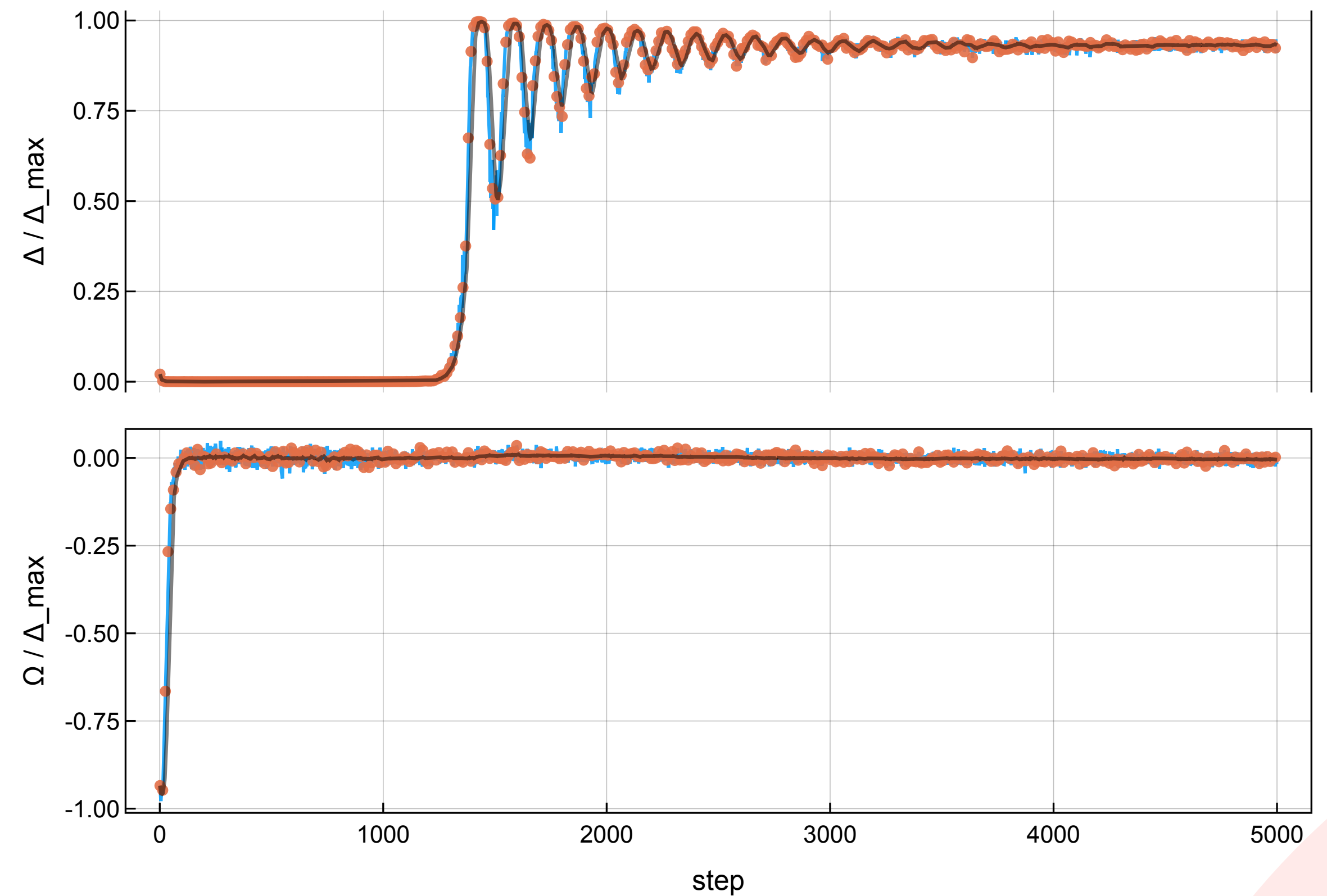
Deep Reinforcement Learning (DRL) Agent-Environment interaction loop



Fock state 1



$$\mathcal{H}(t) = \frac{\Delta(t)}{2} \sigma_z + g(a^\dagger \sigma_- + a \sigma_+) + \Omega_R(t) \sigma_x, \quad \Delta = \omega_q - \omega_m$$



Methods

AI paradigms for quantum control

Supervised Learning (SL) & Regression
training on large datasets

Deep Reinforcement Learning (DRL)
Agent-Environment interaction loop

Evolutionary & Genetic Algorithms
Genetic Algorithms (GA) or Particle Swarm Optimisation

Physics-Informed Neural Networks (PINNs)
integrating the Schrödinger or Lindblad equations into the Neural Network's loss function

AI paradigms for quantum control

Methods

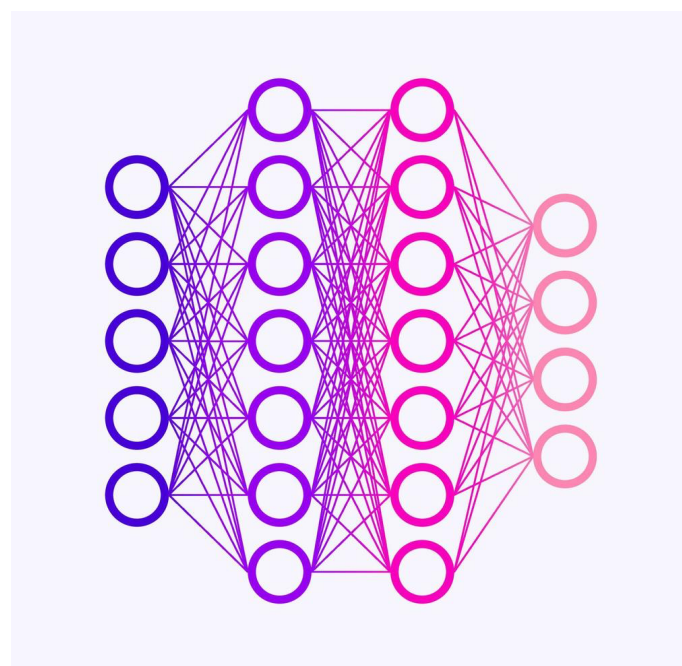
Evolutionary & Genetic Algorithms

Genetic Algorithms (GA) or Particle Swarm Optimisation

Deep Evolution Strategies algorithm

No analytical gradient of the Master Equation is required: **Black Box optimisation**

It's considered Gradient-Free because it uses a **Stochastic Gradient Estimate**. The gradient is not calculated from the physics; it is inferred from the fitness f of random perturbations.



$$\theta_i = \hat{\theta} + \sigma \epsilon_i \quad \epsilon_i \sim \mathcal{N}(0, I)$$

Gaussian noise is added to the Neural Network weights for the exploration

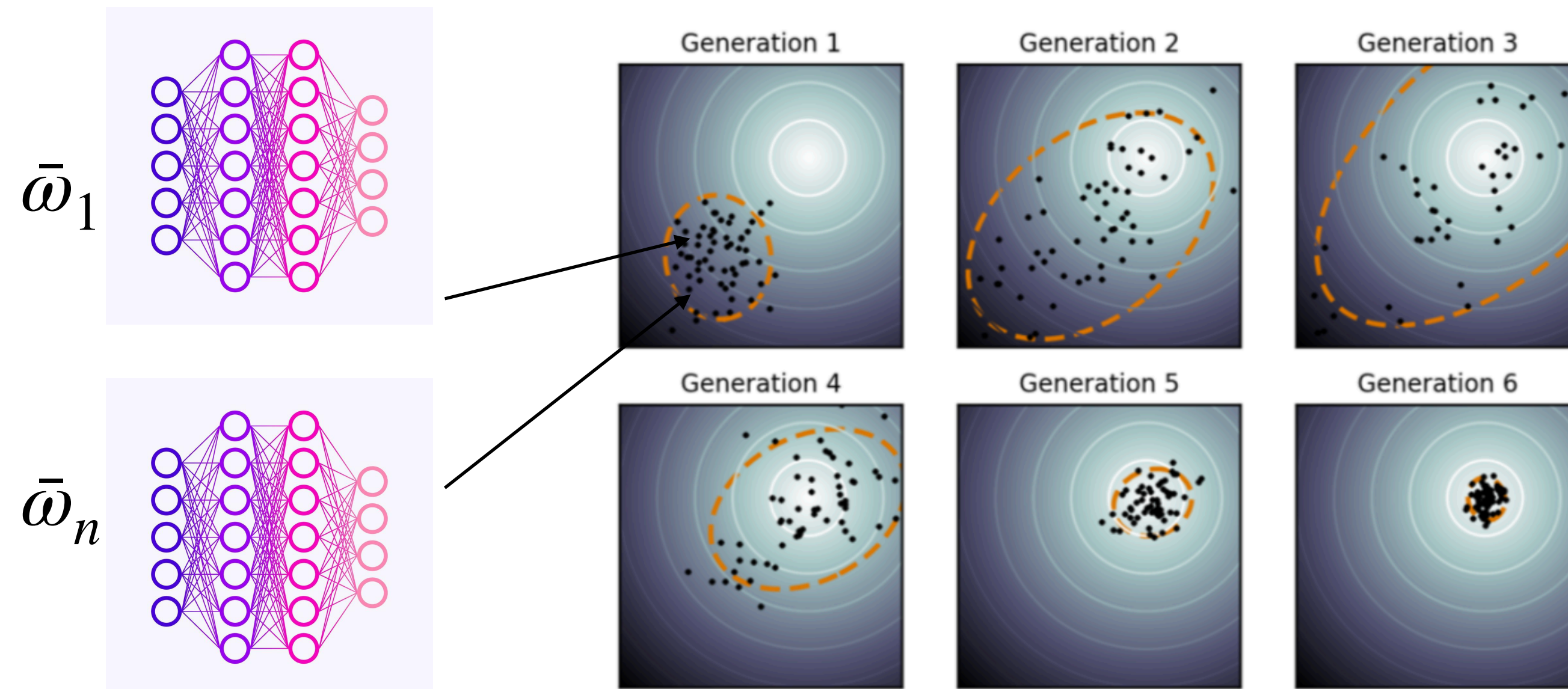
AI paradigms for quantum control

Methods

Evolutionary & Genetic Algorithms

Genetic Algorithms (GA) or Particle Swarm Optimisation

Deep Evolution Strategies algorithm



$$\theta_i = \hat{\theta} + \sigma \epsilon_i \quad \epsilon_i \sim \mathcal{N}(0, I)$$

$$\nabla_{\theta} J \approx \frac{1}{2n\sigma} \sum [f(\theta + \sigma \epsilon) - f(\theta - \sigma \epsilon)] \epsilon$$

$$\theta \leftarrow \theta + \eta \nabla_{\theta} J(\theta)$$

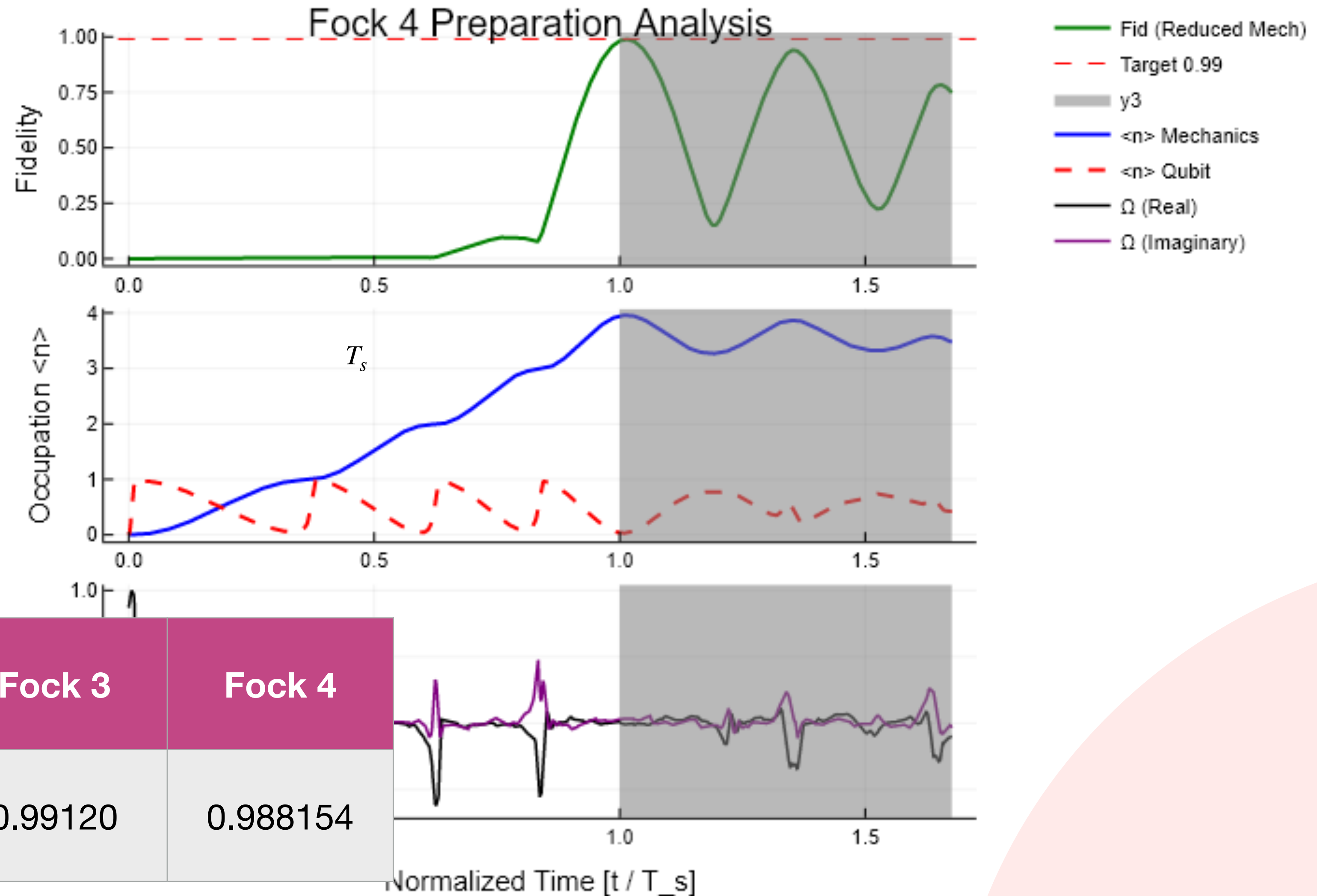
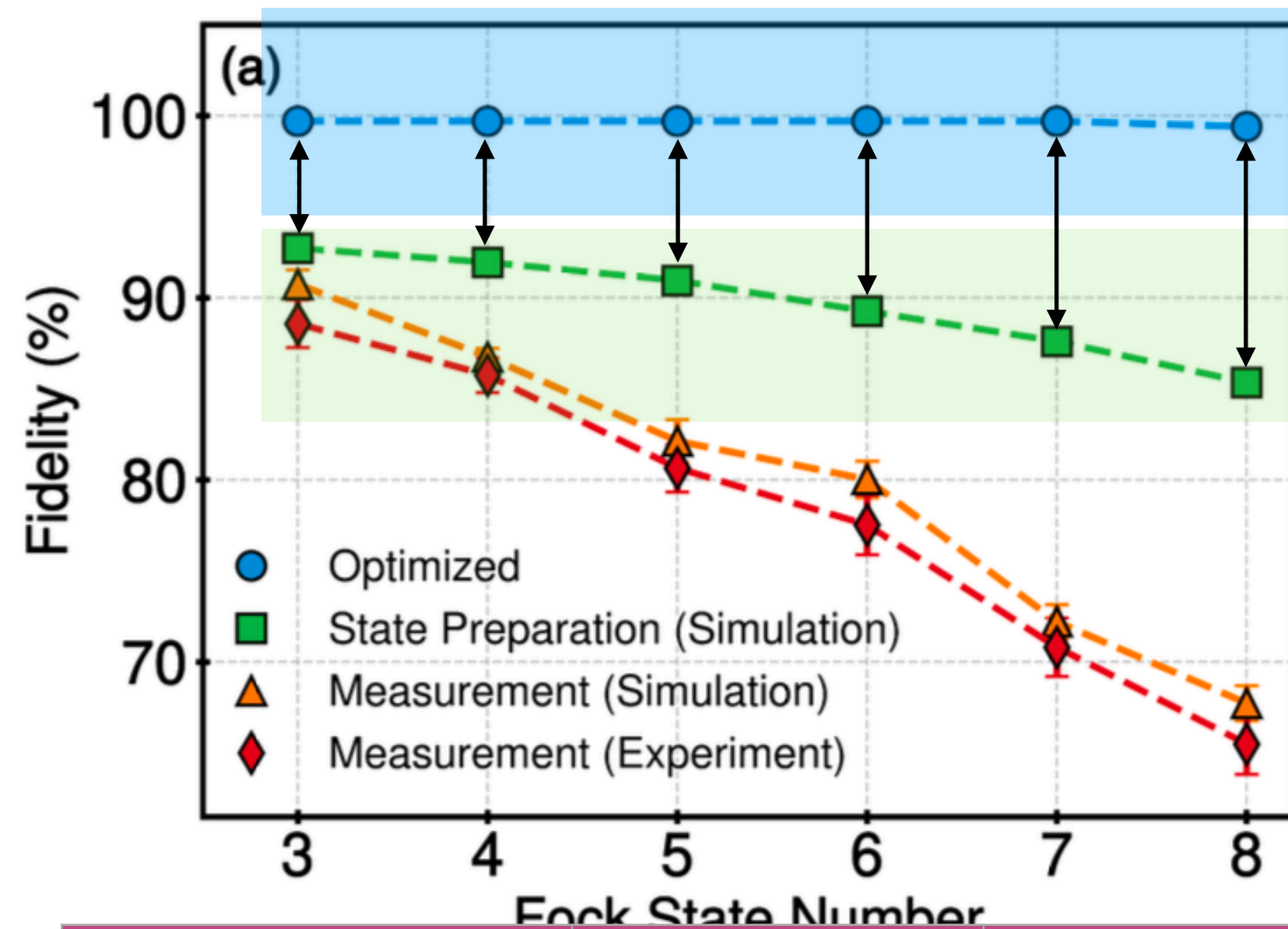
$$f = -\log_{10}(1 - \mathcal{F} + \epsilon)$$

Results

AI paradigms for quantum control

Evolutionary & Genetic Algorithms

FOCK-STATE PREPARATION EXPERIMENT



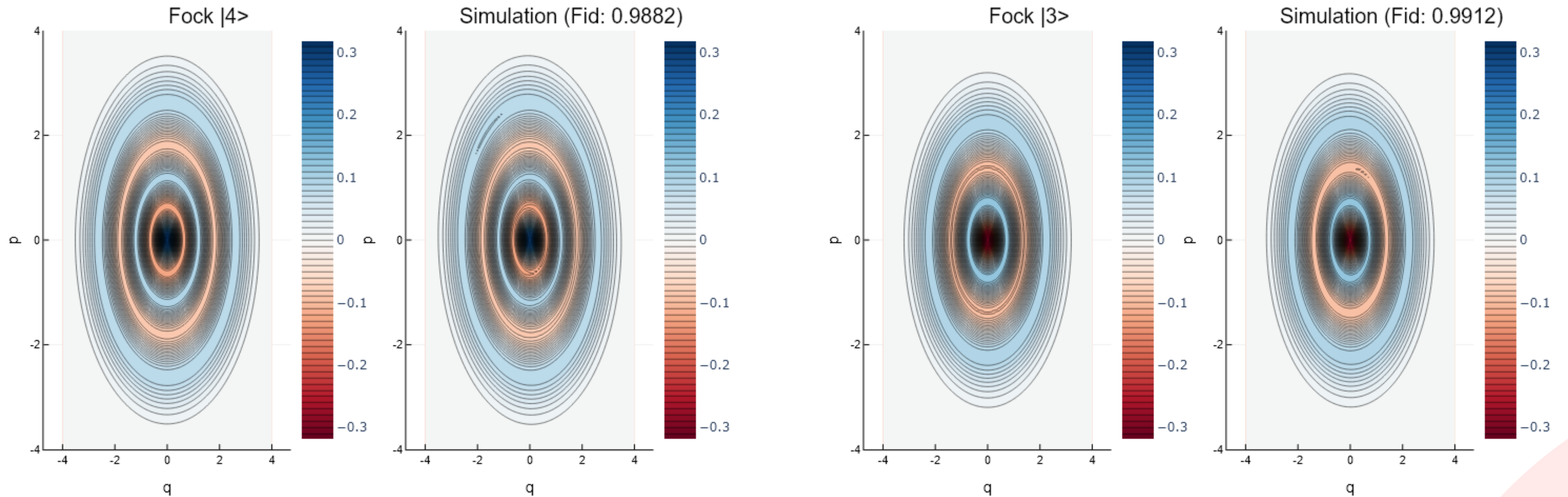
Fock number	Fock 1	Fock 2	Fock 3	Fock 4
ES Fidelity	0.996884	0.99434	0.99120	0.988154

Results

AI paradigms for quantum control

Evolutionary & Genetic Algorithms

Wigner Functions

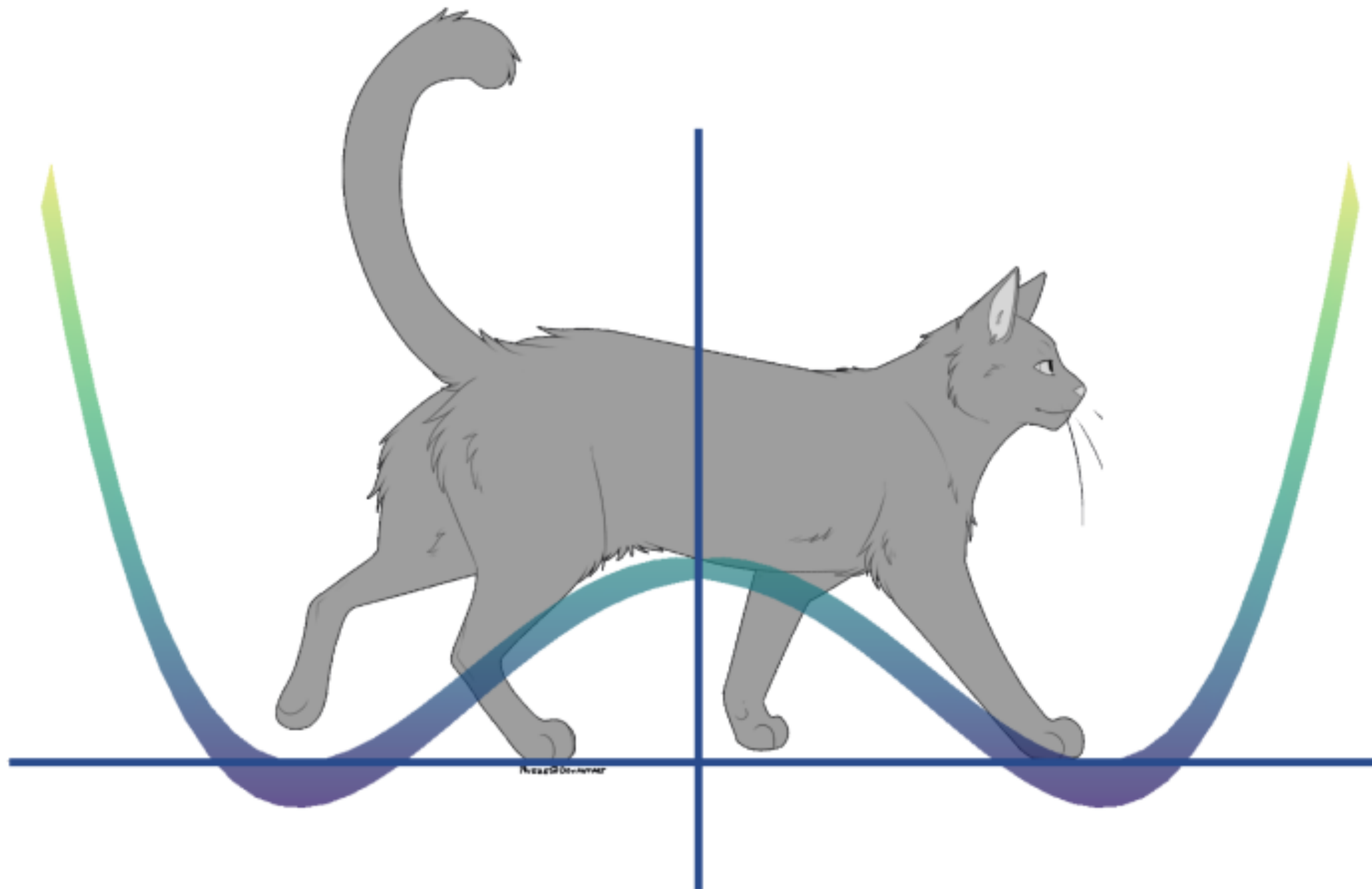


Problems

1. Non-classical states preparation
via AI-based quantum control and
by accounting for non-unitary evolution

2. Mitigating dissipation ?





Thank you!